YASHASVI NAGAR

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SUMMARY

I am a graduate student with hands-on experience in building AI-driven systems and data solutions. I am skilled in machine learning, knowledge graphs, and developing data-driven applications to solve complex problems. Experienced in working with large datasets, developing predictive models, and collaborating in Agile environments to deliver scalable AI and analytics solutions. I am interested in applying AI and data science to support business transformation and deliver impactful solutions.

EDUCATION

Northeastern University ***Boston, MA***

Master of Science in Information Systems *Expected December 2026*

Related Courses: Data Management and Database Design, Designing Advanced Data Architectures for Business Intelligence, Data Science Engineering Methods and Tools, Natural Language Processing Methods and Tools, and Knowledge Graph with GenAI and LLM

Acropolis Institute of Technology and Research ***Indore, MP, India***

B. Tech in Computer Science and Engineering *June 2024*

Related Courses: Database Management System, Data Structure and Algorithm, and Machine Learning

SKILLS

***Programming*:** Python, SQL, R, Java
***Data Engineering & Cloud Tools*:** Microsoft Azure, Snowflake, Databricks
Databases: Neo4j, Oracle, SQL Server
***Visualization Tools*:** Tableau, Power BI
***Tools*:** Streamlit, Alteryx Designer, ER/Studio
***Methodologies*:** ETL Pipeline, Agile, EDA

EXPERIENCE

GraphRAG AI Engineer Intern, CareerGPT ***January 5, 2026 – Present***

- Developing GraphRAG-based features for an AI-powered web application to enable context-aware knowledge retrieval.
- Designing and optimizing graph data models to support scalable querying and efficient information linking.
- Collaborating with the Technical Co-Founder on feature design, user experience flows, and system improvements.
- Building, testing, and deploying full-stack AI features while writing clean, maintainable, production-ready code.

Intern, Cummins Turbo Technologies ***January 8, 2024 – February 24, 2024***

- Worked on a live project focused on image processing and computer vision using Python and OpenCV.
- Collaborated with cross-functional teams to enhance real-time processing performance and optimize system efficiency.
- Applied OpenCV libraries to develop object detection and image analysis features.
- Followed Agile methodology for iterative development, sprint planning, and regular project updates.

PROJECTS

CrimeGraphRAG - AI-Powered Crime Investigation System ***Fall 2025***

- Built an AI-powered investigation assistant using Neo4j and GraphRAG, enabling natural language crime data querying with reduced hallucination.
- Built a complex knowledge graph modeling crimes, suspects, locations, organizations, and evidence.
- Implemented network visualizations using NetworkX and Streamlit, enabling exploration of criminal connections and patterns.
- Designed a conversational AI dashboard with analytics, real-time crime stats, and role-based access control for detectives, analysts, and supervisors.

Emotion Recognition in Multi-Turn Conversations ***Summer 2025***

- Developed context-aware emotion detection system using MELD dataset (13,708 utterances) for 7-emotion classification.
- Implemented BERT baseline, BERT with self-attention, and DialogueRNN models, achieving 70.4% accuracy.
- Addressed class imbalance with weighted loss functions and multi-speaker context using GRU architectures.
- Developed a Streamlit web app featuring real-time predictions, along with trend analysis and pattern recognition in time-series conversation data.

Seattle Pet License Analytics ***Fall 2025***

- Designed a Star Schema data warehouse in Snowflake to track pet licensing trends across Seattle neighborhoods.
- Created fact and dimension tables (Date, Location, Breed) enabling analysis by geography, time periods, and pet characteristics.
- Built ADF pipeline to load CSV data through Parquet into Snowflake dimensional model with MERGE operations.
- Implemented dynamic data flows that automatically adapt to changing data structures, reducing maintenance by 40%.